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Corncob biochar supplementation improves nutrient digestibility, fattening performance and carcass characteristics of fattening sheep

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Abstract

Background: Corncob is a cheap source for biochar in developing countries. No studies reported on use of biochar as sheep fattening supplement.

Objectives: The goal of this study is to determine the effect of corncob biochar supplementation on growth performance and carcass characteristics of sheep.

Methods: Twenty-four Horro yearling rams (18.6 ± 0.938 kg live weight) were randomly distributed to four groups, control (300 g/day concentrate + local hay ad libitum), BC1.5 (control +1.5 g/day biochar), BC3 (control +3 g/day biochar) and BC4.5 (control +4.5/day biochar). The experimental concentrate consisted of 200 g/day wheat bran and 100 g/day Noug cake. The study consisted of a growth trial of 111 days followed by a digestibility trial of 10 days. Carcass characteristics for all rams were recorded at the end of the trial.

Results: Dry matter intake, crude protein intake, dry matter digestibility and acid detergent fibre digestibility of BC1.5 were significantly higher than the control. Corncob biochar supplementation significantly improved daily weight gain and feed conversion ratio of the rams. Supplementation sheep with corncob biochar improved significantly dressing percentage, hindquarters weight, forequarters weight, ribs weight, brisket weigh and rib eye muscle area. There was a significant effect of the dietary treatment on weight of internal organs; however, they were all within the normal range of sheep.

Conclusions: Corncob biochar supplementation would be recommended at a level of 1.5 g/day to improve growth performance and carcass characteristics of sheep.

KEYWORDS
biochar, maize, sheep, supplementation

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1 | INTRODUCTION

The increase in population and income in the developing countries resulted in an increase in demand for animal products. Livestock sector in the developing countries is unable to meet these increasing needs due to the shortage in feed resources in quality and quantity. Accordingly, it is mandatory to increase the efficiency of utilization of the current feed resources.

The agriculture sector is a substantial contributor to global warming by being a main source of greenhouse gas emission. Soil is key source of greenhouse gas emission which includes carbon dioxide (CO_2), nitrogen oxide (N_2O) and methane (CH_4) (Oertel et al., 2016) (Ayaz et al., 2021). Enteric fermentation of feed in rumen is another source of greenhouse gas emission.

Biochar is a product obtained by pyrolysis of organic biomass, such as wood, manure, leaves and organic wastes (Man et al., 2021). Biochar has the potential to mitigate the impact of agriculture on greenhouse gas emissions in many ways. Adding biochar to soil sequesters carbon into the land and suppresses soil-based emissions of methane and nitrous oxide (Gurwick et al., 2013).

Techniques for biochar incorporation into agricultural soils have been researched in detail (Gurwick et al., 2013). The most feasible method is to cultivate biochar into topsoil. However, this method increases the risk of soil erosion (Lehmann, 2012) and is not applicable in the case of natural pastures. Animals could serve as biological vehicle to transport biochar into the cropping soil and natural pastures (Calvelo Pereira et al., 2011). In addition to that, supplementing diets of calves with biochar improve growth and feed efficiency while they decrease methane emission (Gurwick et al., 2013) (Phongphanith & Preston, 2016) (Saroeun et al., 2018) (Leng, 2018).

In goat, biochar was used to improve digestibility, growth, nitrogen retention and to decrease methane production (Silivong & Preston, 2015; Winders et al., 2018). Moreover, biochar was used to remove toxins and pathogens in livestock diet (Man et al., 2021). On the contrary, neutral effect of biochar supplementation was reported in many studies. Growth performance of pigs was not significantly affected by biochar supplementation (Kupper et al., 2015). Biochar addition did not affect weight gain or feed conversion ratio of broilers (Evans et al., 2017). Growth performance and feed conversion ratio of growing ducks were not significantly affected by biochar supplementation (Islam et al., 2014). Biochar produced from pine wood improved nutrient digestibility and microbial protein synthesis in cattle (Saleem, Ribeiro, Yang, et al., 2018). Rice husk-based biochar improved cattle growth which added at a level of 0.06% of the diet (Leng et al., 2012). Rice biochar supplementation at 1% of cattle diet improved growth rate and feed conversion ratio (Phongphanith & Preston, 2016). Rice husk biochar supplementation improved growth and feed conversion ratio of cattle fed on urea–molasses blocks (Saroeun et al., 2018). Ethiopia has large area of natural pasture of 20 million h (FAOSTAT, 2019). Direct cultivation of these large areas with biochar is challenging. Ethiopia has big sheep heard with 31,849,000 heads (FAOSTAT, 2019), which are basically fed on natural pasture. Thus, sheep could be used to distribute biochar in the Ethiopian natural pasture provided that biochar did not have negative impact of growth performance and health of sheep.

To our knowledge, no studies evaluated the effect of biochar supplementation on fattening performance of sheep. Thus, the goal of this study is to determine the effect of inclusion of biochar in fattening sheep on digestibility, growth performance and carcass characteristics.

2 | MATERIAL AND METHODS

2.1 | Sheep, diets and housing

The study procedures were reviewed and approved and recorded under the number of ANSC1157/2022. Corncob biochar was prepared in Bako Agricultural Engineering Research Centre, Bako, Ethiopia (37°03′E, 9°08′N, altitude of 1650 m.a.s.l). Corncob was obtained from the local market in Bako then pyrolysed in a carbonizing tank at temperature of 400°C for 3 h in a low oxygen atmosphere. Corncob biochar was ground to pass a 1-mm mesh then stored in a plastic bag until the experimental diets were prepared. Twenty-four Horro rams (18.6 ± 0.938 kg live weight and on year of age) were randomly distributed to four dietary treatment groups with six rams each, control (300 g concentrate + native hay ad libitum), BC1.5 (control +1.5 g/day biochar), BC3 (control +3 g/day biochar) and BC4.5 (control +4.5 g/day biochar). All animals in BC1.5, BC3 and BC4.5 received biochar daily throughout the whole of the trial.

All rams were fed on 300 g of concentrate mixture (200 g wheat bran and 100 g Noug cake) and had free access to hay prepared from a mixture native grasses (mainly *Cynodon*, *Digitaria* and *Sporobolus*). Noug seed cake is an agro-industrial by-product resulted from oil extraction from Noug seed (*Guizotia abyssinica*). It is commonly used as protein source in livestock feeding in Ethiopia.

Native hay was obtained from Ethiopian Evangelical church Mekane Yesus, chopped to a theoretical length of 5 cm and mixed properly. The ground biochar was mixed with the daily concentrate allowance of every sheep. Hay and concentrate were feed in separate troughs. All experimental sheep had free access to clean drinking water and salt licks. Sheep were drenched with Ivermectin (200 mcg/kg live weight) to control common parasites, vaccinated against common diseases of fattening sheep in Ethiopia (anthrax, pasteurellosis and enterotoxaemia) and then acclimated to the pens and diets for 21 days. The experimental sheep were housed in individual pens (1.5 m × 0.75 m) in an open-sided barn and fed once a day at 08:00 h. The study consisted of a growth trial of 90 days and a digestibility trial of 10 days. The trial consisted of 7 days of total faecal collection after a growth trial of 90 days. Live weight of sheep was recorded before the morning feeding at the start of the trial and every 10 days to estimate daily weight gain. After the end of the growth trial, all sheep were fitted with faecal bags and allowed for 7 days of adaptation and 7 days of total faecal collection. During faecal collection period, the faecal bags were emptied daily, thoroughly mixed, the 10% of the faeces was sampled, and the faecal samples were stored at −20°C until analysed. Residual feeds were collected before morning feeding, weighed on daily basis sampled and then stored in a

fridge at −20°C for chemical analyses. At the end of the feeding trial, all sheep from each treatment were fasted overnight, weighed and slaughtered. Carcass data was recorded for each sheep, including hot carcass weight, dressing percentage (based on slaughter weight), hindquarters weight, forequarters weight, ribs weight, brisket weight, rib eye muscle area, liver weight, kidney weight, heart weight, small intestine weight, large intestine weight, spleen weight, blood weight, pancreas weight, testes weight and rumen weight. The internal organs were expressed as a percentage of live weight.

2.2 | Chemical analysis

All samples of feed, feed refusal and faeces were dried at 105°C overnight in a forced air oven to determine dry matter ((AOAC, 2006); method 934.01). Ash was determined using a muffle furnace at 550°C overnight ((AOAC, 2006); method 942.05). The nitrogen and ether extract were analysed according to Kjeldahl ((AOAC, 2006); method 954.01) and Soxhlet method ((AOAC, 2006); method 920.39), respectively. Crude protein content was calculated by multiplying nitrogen content by 6.26. Neutral detergent fibre was determined according to Van Soest et al. (1991) without use of an alpha amylase and expressed as residual ash exclusive. Acid detergent fibre was presented exclusive of residual ash, and lignin was identified by solubilization of cellulose with sulphuric acid. Biochar was analysed for carbon by dry combustion using a Leco CHN1000 analyser and coal reference material (Leco Corp., St. Joseph, MI, USA). Briefly, 0.1 g of biochar was mixed with 10 ml of deionized water and shaken at 250 rpm for 24 h, and the pH of the supernatant was measured by a pH meter. The Brunauer–Emmett–Teller (BET) surface area of the biochar was determined by a Tristar II 3020 (Micromeritics).

2.3 | Statistical analysis

The experimental animals were blocked (eight animals/block) according to the live weight. Data obtained from the experiment was subjected to an analysis of variance to determine the effect of the dietary treatment on fattening performance and carcass characteristics according to the following model:

$$Y_{ij} = \mu + \mathrm{TRT}_i\,(\alpha i) + B_j + \varepsilon_{ij}$$

where Y_{ij} is response variable, μ is overall mean, αi is the effect of treatment, B_j is the effect of the block, and ε_{ij} is the residual. The treatment means were separated using Fisher's least significance difference test at $p \leq 0.05$. Polynomial contrasts were used to determine whether there was a linear, quadratic and/or cubic relation between the dietary treatment and response variables. Orthogonal polynomials were decomposed in linear, quadratic and cubic functions using the PROC MIXED procedure in SAS. Data analysis was carried out using (SAS, 2012).

3 | RESULTS

Chemical composition of the experimental feeds and corncob biochar is presented in Table 1. Surface area, pH and carbon content of biochar were 50.2 m^2/g, 8.93% and 76.8%, respectively. Nutrient intake of sheep supplemented with different levels of corncob biochar is shown in Table 2. Corncob biochar supplementation improved significantly dry matter intake of hay and the overall dry mater intake compared to the control ($p < 0.001$). Only BC1.5 had significantly higher crude protein intake compared to the control ($p < 0.001$). The daily intake of organic matter, neutral detergent fibre and acid detergent fibre significantly improved due to the dietary treatment ($p < 0.01$); however, BC1.5 had the highest improvement. The difference between BC3 and BC4.5 in the intake of organic matter, neutral detergent fibre and acid detergent fibre was insignificant ($p > 0.05$).

Corncob biochar supplementation improved significantly the digestibility of dry matter, organic matter and acid detergent fibre ($p < 0.001$). However, the increase in the supplementation level did not significantly improve the digestibility of dry matter, organic matter or acid detergent fibre.

The effect of corncob biochar supplementation on growth performance of sheep is presented in Table 3. Corncob biochar supplementation improved significantly the final live weight and the daily weight gain of the experimental sheep ($p < 0.001$). However, BC1.5 had significantly the highest final live weight and daily gain among the dietary treatments. Feed conversion ratio of the rams decreased significantly due to corncob biochar inclusion ($p < 0.001$); however, the improvement in feed conversion ratio was insignificantly different among all inclusion levels.

Table 4 presents the effect of corncob biochar supplementation on carcass characteristics of the experimental sheep. Sheep in BC1.5 had significantly higher hot carcass weight compared to the control ($p < 0.001$). Corncob biochar supplementation improved significantly the dressing percentage of sheep ($p < 0.001$) with the highest improvement found in BC1.5 and BC4.5. Hindquarters weight and forequarters weight of sheep were positively and significantly affected by biochar supplementation ($p < 0.001$). Rams in BC1.5 had significantly the highest brisket weight and rib eye muscle area among all dietary treatments ($p < 0.001$). Liver weight (% live weight) in corncob biochar supplemented diets was significantly higher than that of the control ranging from 1.46% of live weight (control) to 2.09% of live weight (BC1.5) ($p < 0.001$). Kidney and heart weight were not significantly affected by the dietary treatment ($p > 0.05$). The small intestine of rams in BC4.5 was significantly higher than the control ($p < 0.001$). The small intestine of the experimental rams ranged from 1.69% of live weight to 2.2% of live weight. Corncob biochar supplementation increased significantly large intestine weight of sheep compared to the control ($p < 0.001$). Spleen weight was not significantly affected by corncob biochar supplementation ($p > 0.05$). Pancreas weight in BC1.5 and BC3 was significantly higher than the control ($p < 0.001$). The visual examination of all internal organs of the experimental animals confirms that there were no tissue damages.

TABLE 1 Nutrient composition of experimental diets (g/kg dry matter)

	Hay	Wheat bran	Noug seed cake	Corncob biochar
Dry matter	951	955	949	957
Ash	137	54	92	232
Crude protein	82	179	315	52.4
Neutral detergent fibre	819	525	528	804
Acid detergent fibre	502	273	349	510
Acid detergent lignin	292	41	136	165
Ether extract	8.8	35.1	56.3	5.7

TABLE 2 Dry matter and nutrient intake and apparent digestibility of rams supplemented with corncob biochar

			Diets[a]				P		
Intake (g/day)	Control	BC1.5	BC3	BC4.5	SEM [b]	Sig level	Linear	Quadratic	Cubic
Dry matter									
Hay	448c	569a	501b	490b	5.68	<0.001	0.071	0.457	0.147
Concentrate	300	300	300	300					
Total	748c	869a	801b	790b	4.91	<0.001	0.254	0.111	0.325
							0.124	0.258	0.698
Crude protein	107b	117a	108b	107b	0.49	<0.001	0.114	0.211	0.2
Organic matter	667c	772a	713b	703b	4.91	<0.001	0.258	0.27	0.693
Neutral detergent fibre	557c	656a	599b	587b	4.65	<0.001	0.325	0.58	0.547
Acid detergent fibre	282c	343a	308b	302b	2.9	<0.001	0.078	0.69	0.879
Apparent digestibility (%)									
Dry matter	65.2b	72.2a	72a	71.3a	1.2	<0.001	0.06	0.526	0.369
Organic matter	67.7b	74.5a	74.a	73.7a	1.0	<0.001	0.089	0.258	0.08
Crude protein	80.5	83.5	81.7	82.5	1.1	0.442	0.08	0.854	0.087
Neutral detergent fibre	68.2	70.8	70.1	69.2	0.4	0.865	0.078	0.587	0.066
Acid detergent fibre	56.7b	65.3a	62.5a	63.7a	1.3	<0.001	0.58	0.259	0.658

[a]Control = 200 g wheat bran + 100 g Noug seed cake + local hay ad libitum; BC1.5 = control + 1.5 g/day corncob biochar; BC3 = control + 3 g/day corncob biochar; BC4.5 = control + 4.5% biochar. Means in a row with same letters are not significantly ($p > 0.05$).
[b]SEM = standard error mean.

4 | DISCUSSION

Ethiopia has large herd of livestock, including 20,000,000 heads of sheep, 63,000,000 heads of cattle and 34,000,000 heads of goat. Feeding these livestock on the above recommendations would result in distributing 3,100,000 t of biochar annually. Accordingly, the use of livestock as a delivery system could become a novel solution to safely distribute biochar to Ethiopian pastoral soils. Therefore, the current study aimed at identifying the optimum supplementation level of corncob biochar in sheep diet.

The increase in the intake of crude protein, neutral detergent fibre and acid detergent fibre in corncob supplemented sheep is a direct result of the improvement in dry matter intake which is a result of the increase in the rate of passage of feed in the rumen due to the increase in dry matter digestibility. The positive effect of biochar supplementation is in agreement with studies on calves (Gurwick et al., 2013; Leng, 2018; Phongphanith et al., 2016; Saroeun et al., 2018) and goat (Silivong & Preston, 2015; Winders et al., 2018). However, this result disagrees with Terry et al. (2020) who reported on a neutral effect of pine wood biochar on steers. Biochar produced from wood did not affect growth performance of pigs (Kupper et al., 2015). Poultry litter biochar did not improve growth performance of broilers (Evans et al., 2017). Growth performance and feed conversion ratio of growing ducks were not significantly affected by charcoal (Islam et al., 2014). The disagreement among these studies could be related to the variability of biochar efficacy because of the differences in biochar particle size (McFarlane et al., 2017), origin and preparation process (Waheed et al., 2013).

The increase in acid detergent fibre intake and digestibility indicate to an improvement in energy intake as acid detergent fibre is closely

TABLE 3 Growth performance of rams fed diet supplemented with corncob biochar

	Diets[a]				*p*				
	Control	BC1.5	BC3	BC4.5	SEM[b]	Sig level	Linear	Quadratic	Cubic
Initial body weight (kg)	18.5	19.0	18.6	18.3	0.97	0.716	–	–	–
Final body weight (kg)	22.1b	25.2a	23.4b	23.5b	0.98	<0.001	0.5	0.369	0.741
Average daily gain (g/day)	41.9c	67.6a	53.5b	56.8b	3.47	<0.001	0.258	0.147	0.254
Feed conversion ratio	17.9a	12.8b	14.9b	13.8b	0.39	<0.001	0.698	0.658	0.2

[a]Control = 200 g wheat bran + 100 g Noug seed cake + local hay ad libitum; BC1.5 = control + 1.5 g/day corncob biochar; BC3 = control + 3 g/day corncob biochar; BC4.5 = control + 4.5% biochar. Means in a row with same letters are not significantly ($p > 0.05$).
[b]SEM = standard error mean.

related with dry matter digestibility of forage in ruminants (Horrocks & Vallentine, 1999).

The positive effect of biochar supplementation on growth performance could be explained through different mechanisms. Adding biochar to the diet improved microbial protein synthesis in sheep rumen (Mirheidari et al., 2020) which would improve amino acids supply to the small intestine. The microbial production of acetate, propionate and total volatile fatty acids in the rumen increased by biochar supplementation of the diet (Saleem, Ribeiro, Alipour, et al., 2018).

Furthermore, intestinal villus height increased by adding biochar in the diet (Ruttanavut et al., 2009) which would increase nutrient absorption to the blood stream. Biochar inclusion and the diet could decrease methane production in rumen leading to better energy utilization (Man et al., 2021). Biochar could adsorb bacteriological pathogens and their metabolites (Schmidt et al., 2019) which could improve the overall function of rumen and general health leading to improvement in the overall growth performance.

The improvement in the final live weight of the biochar supplemented sheep might explain the improvement in hot carcass weight. The improvement in dressing percentage, hindquarters weight, fore quarters weight, ribs weight, brisket weight and rib eye muscle reported as a result of biochar supplementation is in-line with Dixon et al. (2000), Ponnampalam et al. (2002) and Turner et al. (2014) who reported on an increase in carcass traits as a result of the elevation in nutrients supply. Biochar supplementation could improve skeletal muscles development (Rashidi et al., 2020) which contributed to the overall improvement in carcass traits. The improvement in carcass traits means that corncob biochar supplementation of sheep will produce more edible parts.

Although there was a significant effect of the dietary treatment on the relative sizes of internal organs of the experimental sheep, the size of these organs fall within the normal variability which is reported for sheep (Girma, 2010); (Tefa, 2006). Furthermore, no damage was reported in any of these organs. Thus, inclusion of corncob biochar up to 4.5 g/day is not associated with any health problems in Horro sheep.

Biochar is a very efficient tool to decrease GHG emission from both soil and ruminant animals. However, direct application biochar to the soil of cropping land and natural pasture is restricted by many practical limits. The current study shows that inclusion of corncob biochar in sheep improved fattening performance and carcass traits without any negative effect on health. Accordingly, inclusion of corncob biochar in sheep diets would have multiple advantages in the mixed farming systems. The biochar consumed by sheep would not be absorbed and is expected to be excreted with the manure. This manure could be transported to cropping plots as an organic fertilizer or to natural pasture. Accordingly, biochar would be integrated into the soil then decreasing GHG emission.

In conclusion, use of corncob biochar as feed supplementation at 1.5 g/day improved feed intake, nutrients digestibility, growth, feed conversion ratio and carcass characteristics of sheep.

TABLE 4 Carcass characteristics of ram supplemented with corncob biochar

	Diets[a]					*p*			
	Control	BC1.5	BC3	BC4.5	SEM[b]	Sig level	Linear	Quadratic	Cubic
Hot carcass weight (kg)	8.04b	10.3a	9.08ab	9.37ab	0.17	<0.001	0.2	0.617	0.23
Dressing percentage[c]	36.4c	40.8a	38.9b	40a	0.29	<0.001	0.068	0.821	0.212
Hindquarters (kg)	2.11b	3.19a	2.77a	2.81a	0.1	<0.001	0.098	0.078	0.362
Forequarters (kg)	1.72b	2.29a	2.09a	2.11a	0.04	<0.001	0.457	0.164	0.33
Ribs (kg)	1.06c	1.4a	1.26b	1.32ab	0.02	<0.001	0.444	0.626	0.745
Brisket (kg)	0.5c	0.7a	0.61b	0.6b	0.005	<0.001	0.547	0.66	0.367
Rib eye muscle area (cm^2)	9.28c	12.6a	11.4b	11.7b	0.14	<0.001	0.093	0.654	0.254
Organs (% live weight)									
Liver	1.46b	2.09a	1.96a	1.94a	0.073	<0.001	0.847	0.9	0.39
Kidney	0.317	0.339	0.312	0.291	0.022	0.481	0.258	0.325	0.41
Heart	0.602	0.622	0.631	0.614	0.028	0.615	0.256	0.974	0.47
Small intestine	1.69b	1.93ab	1.87ab	2.12a	0.088	<0.001	0.569	0.397	0.36
Large intestine	1.25c	1.52b	1.51b	1.72a	0.053	<0.001	0.0547	0.617	0.366
Spleen	0.255	0.29	0.283	0.293	0.017	0.149	0.0698	0.761	0.467
Pancreas	0.232c	0.3a	0.273ab	0.245bc	0.013	<0.001	0.0789	0.754	0.671
Omasum and abomasum	0.752	0.84	0.774	0.784	0.027	<0.001	0.211	0.657	0.364
Rumen and reticulum	1.70b	2.18a	2.04ab	2.26a	0.117	<0.001	0.369	0.745	0.397

[a]Control = 200 g wheat bran + 100 g Noug seed cake + local hay ad libitum; BC1.5 = control + 1.5 g/day corncob biochar; BC3 = control + 3 g/day corncob biochar; BC4.5 = control + 4.5% biochar; Means in a row with same letters are not significantly ($p > 0.05$).
[b]SEM = standard error mean.
[c]Slaughter weight basis.

AUTHOR CONTRIBUTIONS

Conceptualization; formal analysis; funding acquisition; investigation; methodology; project administration; resources; supervision; validation; writing—original draft; writing—review and editing: Solomon Demeke. *Formal analysis; writing—original draft; writing—review and editing*: Ashraf Alkhtib.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data of the current study is available from the corresponding author on reasonable request.

ETHICS STATEMENT

This study was approved by the Research and Ethical Review Board (RERB/IRB) of JUCAVM, Jimma University (ANSC1157/2022).

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PEER REVIEW

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